

Chapter 9

Estimated Casualty Risk for Disaster Preparation in Five Scenario Great Earthquakes, Sichuan-Yunnan Region, China



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Abstract For earthquake disaster preparation and risk reduction, quantitative estimates of casualties caused by potential future ruptures is crucial. We estimate the numbers of casualties and the population strongly affected for five “worst-case scenarios” of devastating earthquakes in the Sichuan-Yunnan region, China. These earthquake scenarios are assumed to rupture the entire segments of the Xianshuihe, Anninghe, Zemuhe, Daliangshan and Xiaojiang faults. The source parameters of the scenario earthquakes are derived from fault models. Before calculating scenario losses however, we validate the algorithms and data sets of Quake Loss Assessment for Response and Mitigation (QLARM) for the 13 earthquakes of the mainland of China since 2003, which have caused fatalities, by matching observed intensities and casualty counts with calculated values. These near real-time fatality estimates are reviewed and found to have been correct and, for people injured, six are 100% correct and three have been within a factor of 2.5. We use an average attenuation and assume that earthquakes occur at 03:00 local time, which expects assumes maximum occupancy of buildings. The greatest disaster is derived from the scenarios of the Xiaojiang fault rupture, where we calculate the possible numbers of fatalities on average as 286,000 and injured as 751,000. The scenario of the maximum rupture along the Daliangshan fault results in approximately twice the casualties reported in

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the 2008 M_W 7.9 Wenchuan earthquake. The three scenarios along the Xianshuihe, the Anninghe and the Zemuhe faults result in estimates of casualties similar to those in the Wenchuan case. These estimates are significant not only for disaster preparation and relief by the government, but also for risk prevention and mitigation by the public.

Keywords Earthquake casualty risk · Human loss estimates · Sichuan-Yunnan region · China · Earthquake disaster preparation

9.1 Introduction

Quantitative estimates for casualty risk caused by potential future earthquakes are valuable for disaster preparation and risk reduction. The earliest tools for estimating losses due to earthquakes were EXTREMUM (Shakhramanjan et al., 2001) and HAZUS (Federal Emergency Management Agency, 2003), the first of these responded to worldwide events, the second to USA earthquakes only. The tool we used, Quake Loss Assessment for Response and Mitigation (QLARM), is a second generation developed in part based on EXTREMUM, with elements of HAZUS incorporated. A summary of various approaches to earthquake risk estimates is presented by Erdik et al. (2014) and Wyss (2014). Many of these publications address a specific country, a region or a single city. Worldwide near real-time earthquake loss estimates for large events (e.g., $M \geq 5$ for Europe and China) are provided by PAGER since 2006 (Jaiswal & Wald, 2010; Jaiswal et al., 2011) and QLARM since October 2002 (Wyss, 2014).

Near real-time loss estimates are based on earthquakes, that have happened and are deterministic. Estimates of monetary losses for insurance purposes are probabilistic as a concept (e.g. Salgado-Galvez et al., 2016a, 2016b, 2017). The quantitative scenario loss estimates we present here are deterministic, based on assumed ruptures of active faults with histories of large to great earthquakes in the Sichuan and Yunnan provinces of China.

A description of the engine of QLARM can be found in the *Appendix*. Based on QLARM, loss scenarios have been published for the Himalayan plate boundary (Wyss, 2005, 2006), Romania (Trendafiloski et al., 2009), Algeria (Rosset & Wyss, 2017; Rosset et al., 2009), Indonesia (Wyss, 2010), Chile (Wyss, 2010), Mexico (Wyss & Zúñiga, 2016), Nepal (Wyss, 2017; Wyss & Chamlagain, 2018), India (Wyss et al., 2018), Italy (Wyss and Rosset, 2020), Kyrgyzstan (Rosset et al., 2020), the Azores (Fontiela et al., 2020; Wyss et al., 2018), and Colombia (Wyss et al., 2021).

Here, we select five faults with high seismic risk in the Sichuan-Yunnan region, China, and construct “worst-case scenarios” of devastating earthquakes, although even greater earthquakes are possible along these faults. Before the calculation for scenario losses, we validate the algorithms and data sets of the QLARM software using the 13 earthquakes, which have recently caused fatalities on the Chinese mainland, by matching observed intensities and casualty counts with calculated ones.

All our calculations for scenarios are subject to assumptions and are affected by uncertainties, thus they should be considered order of magnitude estimates only. The assumptions we set forward are based on the validation work, and listed below:

- (1) the fault zones are supposed to host the largest ruptures we can expect according to the length of the investigated fault segments;
- (2) the energy radiated is assumed to come from lines with a depth of 15 km;
- (3) we propose to use the attenuation of $C_3 = 3.9$ (refer to Eq. 1 in the *Appendix*);
- (4) the occurrence time is assumed as 03:00 local time, which means maximum occupancy of buildings;
- (5) the losses are calculated for 2020, and the regional population growth rate of the past few years is from the National Bureau of Statistics of China (see *Data and Resources*).

Approximately correct estimates of fatalities in near real-time (minutes after earthquakes) and good matches of earthquake casualties and observed intensities by theoretical values using line sources performed in this paper, give us confidence that our scenario results are realistic.

9.2 Validation of QLARM for the Mainland of China

We wish to validate the algorithms and data sets in QLARM for the mainland of China by using the final parameters for each earthquake for which intensity and fatality information exist since 2003. Theoretical intensities and numbers of casualties for the 13 earthquakes listed in Table 9.1 have been computed for comparison with observed values. Because QLARM is constructed to use the standard USGS M values, we therefore used the M_W from the U.S. Geological Survey (USGS; see *Data and Resources*) for our calculations.

We trust the epicenters provided by the China Earthquake Networks Center (CENC) rather than those by the USGS because they are derived from arrival times in regional networks. In addition, we consider macroseismic information for refining the epicenters. When the instrumental epicenters are offset from the center of the isoseismals, we assume that the difference is due to errors in the instrumental solution and use the latter for loss calculations because it represents the center of energy radiated.

Both of the USGS moment tensor depths and those based on the data from the regional networks are considered. However, depth estimates are uncertain, thus we use intensity observations to refine the depths, unless detailed source models with slip as a function of depth are available. Therefore, the depths shown in Table 9.1 are derived from the best fit of intensities calculated by QLARM matching the reported isoseismals.

Earthquake ruptures can be approximated as point sources when the magnitude is smaller than 6.5. For larger ruptures and cases with elongated isoseismal patterns, we assume that energy radiates from line sources. We estimate the lengths and widths of

Table 9.1 Earthquakes in the mainland of China with near real-time alerts from January 1, 2003 to April 1, 2017

Name	yyyy/mm/dd	hh:mm:ss	Lon	Lat	Depth	M_w	C_3
Dayao	2003/07/21	15:16:31	101.16	25.95	11	6.0	3.5
Zhaosu	2003/12/01	01:38:31	80.47	42.96	21	6.0	4.4
Wenchuan	2008/05/12	06:28:01	104.38	31.90	15	7.9	3.7
Haixi	2009/08/28	01:52:06	95.78	37.64	20	6.3	3.5
Yushu	2010/04/13	23:49:38	96.85	33.11	15	6.9	3.5
Xinyuan-Hejing	2012/06/29	21:07:33	84.69	43.32	25	6.3	4.5
Lushan	2013/04/20	00:02:47	102.97	30.25	26	6.6	4.0
Minxian-Zhangxian	2013/07/21	23:45:56	104.14	34.55	18	5.9	4.5
Ludian	2014/08/03	08:30:13	103.38	27.08	12	6.2	4.1
Jinggu	2014/10/07	13:49:39	100.50	23.37	18	6.1	4.1
Pishan	2015/07/03	01:07:47	78.15	37.56	20	6.4	4.0
Zaduo	2016/10/17	07:14:50	94.96	32.83	15	5.9	3.5
Arketao	2016/11/25	14:24:30	74.18	39.22	19	6.6	3.5

The epicenter coordinates and the depths are those used here for intensity calculations, M_w is the USGS magnitude, C_3 controls the attenuation according to Shebalin (1985). Times are UTC

the rupture from M , using established relations (Wells & Coppersmith, 1994; Wyss, 1979). In Table 9.2 we give the end-points (X_i , Y_i), dip, width (W), length (L) and depth (Z) of ruptures for $M \geq 6.5$ earthquakes in Table 9.1. In the case of line sources, the epicenter used for the calculation is artificially placed at the center of the rupture

Table 9.2 Fault model parameters of the $M \geq 6.5$ earthquakes given in Table 9.1

Name	Epi Lat	Epi Lon	Z	L	W	Dip	Y_1	X_1	Y_2	X_2	From
Wenchuan	31.90	104.38	15	305	25	50	31.10	103.45	32.70	105.31	Zhang et al. (2009)
Yushu	33.11	96.85	15	30	20	80	33.20	96.73	33.02	96.97	Zhang et al. (2010)
Lushan	30.24	102.97	26	25	30	30	30.14	102.93	30.35	103.01	Zhang et al. (2013)
Arketao	39.22	74.18	19	30	10	80	39.27	73.99	39.18	74.38	see Data and Resources

X_i and Y_i are the assumed end-points of the respective line ruptures. The epicenter is placed at the center of the line rupture

to provide symmetrical calculations of intensities. The last column in Table 9.2 lists the detailed source parameters of the fault models.

We match the reported isoseismals by calculated intensities adjusting the depths (Z) and the attenuation (C_3 in the Eq. 1 of *Appendix*) when necessary. The best fitting depth is obtained once the intensity points calculated by QLARM for that depth meet the following criteria:

- (1) $I_{\max}$ (calculated) equals $I_{\max}$ (reported);
- (2) the distributions for calculated intensity ($\geq VI$) points match the reported isoseismals well at high intensities, and within a deviations of 40–60 km in parts of the area shaken less strongly;
- (3) the calculated casualties (fatalities and injured) approximately equal the reported numbers of fatalities and injured (Table 9.3).

We give max, middle and min weights to the criteria (1), (2) and (3), respectively. The number of reported casualties comes from the China Earthquake Administration (CEA) and news media.

The fractions of buildings and population in them used for the three size classes of settlements and the building classes are given in Table 9.4. We have modeled settlements in the mainland of China in three size classes, < 6000 inhabitants, between 6000 and 50,000, and > 50,000 in the data set of QLARM. These fractions of building classes that make up 100% of the buildings in one settlement vary in different countries. The European Macroseismic Scale (EMS-98) defines classes ranging from the weakest (A) to the most resistant building (F). This first order approximation will be refined in the future. At present, we accept this model for buildings and inhabitants, under the condition that the resulting validation results are correct. Although these data can and should be further refined, they are accepted in this study, because the overall validation results are correct. The degree of damage and the human losses are calculated for each building type in a settlement, based on the distributions of building stock and population of Table 9.4 and the intensity predicted using Eq. 1 of the Appendix.

The intensities reported in China are on a different scale from MMI (modified Mercalli intensity) used by QLARM. However, this difference is small (Hu, 2009; Wan, 2016; Yuan & Tian, 2012), thus we deem the Chinese intensity scale is approximately the same as MMI. Furthermore, intensities are usually reported as integers without fractions, so we define bands of intensities calculated by QLARM as rounded to the nearest integer.

In order to render this paper more readable, we explain in detail the validation of the 2003 Dayao earthquake as a point source, and the 2008 Wenchuan earthquake as well as the 2014 Jinggu earthquake as a line sources, but we refer the reader to the section on *Supplementary Material* for the validation of the remaining 9 earthquakes.

Table 9.3 The number of fatalities and injured reported compared to those calculated in near-real time using point sources and those in this paper using line sources for the earthquakes given in Table 9.1

Name	WAPMERR/ICES Foundation		This paper		Reported casualties	
	Fat ¹ (est)	Inj ¹ (est)	Fat ² (est)	Inj ² (est)	Fat ³	Inj ³
Dayao	95–260	–	30–370	150–1,000	16	584
Zhaosu	2–7	7–40	10–110	40–360	10	47
Wenchuan	40,000–100,000	95,730–414,259	55,000–140,000	146,000–378,000	87,149	374,643
Haixi	0–10	0–40	0–30	10–120	0	?
Yushu	200–4,000	500–10,000	450–3,000	2,000–10,000	2,968	10,701
Xinyuan-Hejing	0–50	10–200	30–350	130–1,000	0	52
Lushan	200–2,300	1,000–7,700	210–3,000	1,000–11,000	217	13,484
Minxian-Zhangxian	0–300	100–1,000	130–2,000	640–6,000	95	2,414
Ludian	80–635	316–2,233	190–1,000	650–4,000	617	3,143
Jinggu	0–400	100–1,400	70–1,000	390–4,000	1	331
Pishan	0–200	50–500	50–480	190–2,000	3	260
Zaduo	0–0	0–0	0–10	0–40	1	?
Arketao	0–50	20–170	10–170	60–600	1	?

Table 9.4 Distribution of building stock by EMS-98 classes and city sizes for the region of China studied

Settlement size	Urban (> 50,000 inh.)				Mid-urban (6000–50,000 inh.)				Rural (< 6000 inh.)		
EMS-98 classes	A	B	C	D	A	B	C	D	A	B	C
Building distribution	0.05	0.10	0.70	0.15	0.20	0.65	0.10	0.05	0.5	0.35	0.15
Population distribution	0.05	0.10	0.70	0.15	0.20	0.65	0.10	0.05	0.5	0.35	0.15

9.2.1 The Dayao Earthquake of 21 July 2003

For the 21 July 2003, Dayao earthquake located in the center of Yunnan Province, the magnitude was $M_S 6.2/M_W 6.0$, released by the CEA and USGS, respectively. We used a point source model for this earthquake. The isoseismal map for that earthquake is from Fei et al (2004), with $I_{\max}$ (reported) equal to VIII. The $I_{\max}$, spatial distribution of intensities and the number of casualties match best when using a depth of 11 km (Fig. 9.1). The distance by which we adjusted the epicenter from CENC to the center of the isoseismals is about 7 km.

$I_{\max}$ (calculated) is VIII and the estimated ranges of fatalities and injured are 30–370 and 150–1000, respectively (Table 9.3). The ranges of the estimation of fatalities and injured are the formal limits for the result being within 95% probability. We take this wide range because there are additional uncertainties, such as precise location of the scenario earthquake, precise magnitude M , precise attenuation, and spectral distribution of radiated energy. The calculation for the mean degree of damage and the human losses depends on intensity I as elaborated by Trendafiloski et al. (2011).

The intensity estimates, which depend on human perception, disturbance of objects, damage to buildings/structures and disruptions of the Earth's surface, are determined by macroseismic intensity surveys by the CEA. The site conditions and the type of buildings/structures have a significant influence on the isoseismals. Concerning these parameters, we have no relevant knowledge and cannot simulate the effects.

The model of QLARM that we preferred for the Dayao earthquake balances the difference between the observations and the calculations and makes them acceptable for each parameter. The parameters used and the constraints obeyed in the model we used for this earthquake and the subsequent ones (Table 9.1) are as follows.

- (1) The depth we use is based on the results by CENC and the moment tensor solution of USGS, and is refined by intensity observations, unless seismic detailed source models are available.
- (2) The occupancy rate is computed based on local time.
- (3) The attenuation function used in our calculation is an average for the world, using the constant $C_3 = 3.5$ (Shebalin, 1985; Appendix) when we have no

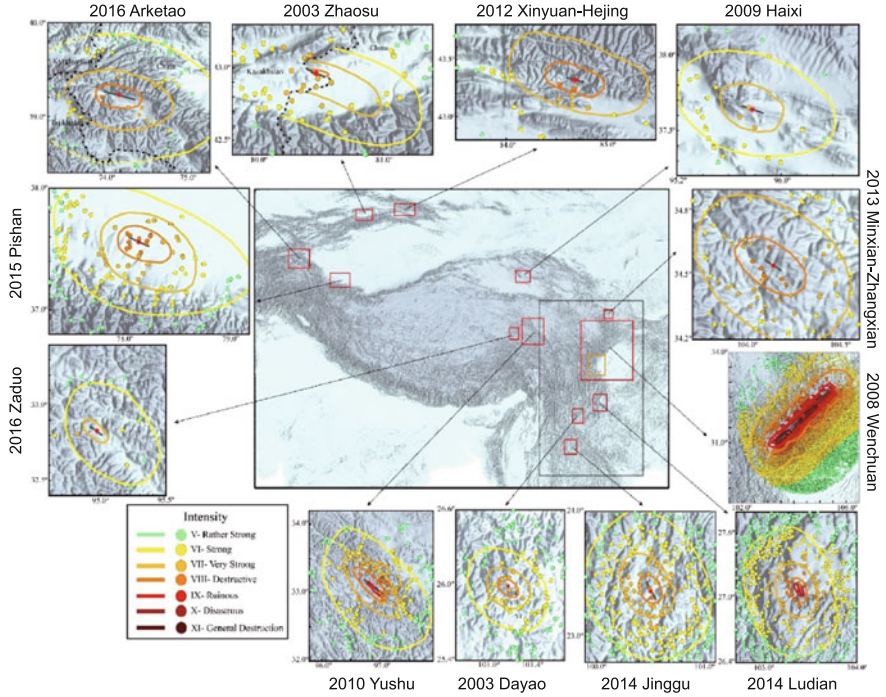


Fig. 9.1 Map of isoseismals and intensities calculated by QLARM (colored dots) for the events in Tables 9.1 and 9.2. The hexagrams indicate the macroscopic epicenters and the gray solid lines depict the line sources we used for calculations. Parameters used by QLARM for calculation are given in Tables 9.1 and 9.2. The yellow rectangle of the indexing figure shows the 2013 Lushan earthquake, in Sichuan province. The black rectangle shows the region of Fig. 9.3

- other information. If the observed distribution of intensities demands it, we have used different factors for C_3 and discuss these in the cases concerned.
- (4) The population data of settlements of QLARM in Sichuan-Yunnan region are from 2010. The number of computed fatalities given is corrected for the respective earthquake dates, using the annual population growth rate for China released by the National Bureau of Statistics of China (see *Data and Resources*).
 - (5) The farther a settlement is from the epicenter, the less reliable and significant is its comparison of reported with calculated intensities. So, for the matches we used mostly the settlements within the fields of high intensities.

9.2.2 The Wenchuan Earthquake of 12 May 2008

For the 12 May 2008, Wenchuan $M_S 8.0/M_W 7.9$ earthquake of Sichuan province, we used the isoseismals (Fig. 9.1) released by the CEA and collected by Chen and Booth

(2011). The isoseismal $I_{\max}$ equals XI and is elongated in the ENE direction. The highest intensities occurred in two separate areas, namely Wenchuan and Beichuan counties. On 25 August 2008, 87,149 people were reported killed (we consider 17,923 missing people to be dead) and at least 374,643 were injured (Table 9.3).

The spatio-temporal rupture process of the Wenchuan earthquake has been studied by several researchers e.g., (Ji, 2008; Wang et al., 2008; Du et al., 2009; Zhang et al., 2009; Huang et al., 2012; Source model of the Wenchuan rupture published by finite-source rupture model database of Source Inversion Validation project, see *Data and Resources*; Source model of this event released by USGS, see *Data and Resources*). However, in order to ensure consistency of the source models with other events used here, we only used results provided by Zhang Y. et al. (2009) for the Wenchuan earthquake.

We used a line source with a length of 305 km of which the end-points are given in Table 9.2. It is placed along the surface break and approximately within the center of the simplified source rectangle, which is based on asperities, and has an overall rupture width of 25 km and a dip of 50°. To evenly cover the area affected by strong shaking, we artificially placed the epicenter at the center of the line source. This artificial location is not the initial rupture point at the southern end.

The $I_{\max}$ and distribution of intensities of QLARM match the reported ones best if we assume a depth of 15 km (Fig. 9.1). The two sides of the rupture (mountains versus plains) have different attenuation, so we need to find a compromise. An attenuation with $C_3 = 3.7$ was used because the fields for IX and VIII are relatively matched within a deviation of 40 km and 60 km in this case. No model that is otherwise reasonable results in intensity XI, so we accept the calculated $I_{\max} = X$, understanding that amplitudes of seismic waves are amplified by soil conditions locally. The ranges of the calculated number of fatalities and injured are 55,000–140,000 and 146,000–378,000, respectively (Table 9.3).

9.2.3 The Jinggu Earthquake of 7 October 2014

For the Jinggu $M_S 6.9/M_W 6.1$ earthquake of 7 October 2014 in Yunnan province, the isoseismal plot is shown in Fig. 9.1, and $I_{\max}$ equals VIII. The moment magnitude and focal mechanism of the Jinggu earthquake are similar to these of the 2014 Ludian earthquake. However, the damage caused by the Ludian and Jinggu earthquakes are different. Only 1 person in Jinggu, but 617 in Ludian were reported dead. The $I_{\max}$ values were not equal in the two cases and the number of people affected in the earthquake with more fatalities is higher.

The difference in damage may also have been due to a difference in the energy radiated (Choy & Boatwright, 2009; Choy et al., 2001; Song et al., 2016). Song et al (2016) reported different magnitudes for the Jinggu and the Ludian earthquakes. It can be seen from their results that although the M_W of the Jinggu earthquake approximately equals that of the Ludian earthquake, there are differences between them in the energy magnitude M_e and the broadband body-wave magnitude m_{Bc} ,

respectively. The differences in both M_c and m_{Bc} between the two earthquakes are about 0.3, meaning that the energy radiated at high frequencies is lower by a factor of 3 in the case of Jinggu earthquake compared to Ludian earthquake (Choy & Boatwright, 1995). This difference in the source spectra can explain partially the difference in the losses.

For the model calculation, we moved the epicenter from CENC to the macroepicenter. The adjusted distance is about 6 km. Owing to the isoseismals being elongated in the ESE direction, we assume that the end-point coordinates of this line source with length of 13 km are 23.44° N, 100.46° E and 23.30° N, 100.55° E. The match is good when we increase the depth of the hypocenter from those of CENC and USGS to 18 km along with $C_3 = 4.1$ (Fig. 9.1).

The calculated ranges of casualties, 70–1000 and 390–4000, respectively (Table 9.3), are overestimated. We propose that weak energy radiation in the Jinggu earthquake, as shown by the magnitudes sensitive to high frequencies, was the main cause of fewer casualties than expected.

9.2.4 Loss Estimation for an Aftershock of the Second Kind: The Lushan Earthquake of 20 April 2013

The epicenter of the Lushan earthquake, which happened about 5 years after the Wenchuan earthquake, was located 90 km southwest of the latter rupture (Wu et al., 2014). On 20 April 2013, the Lushan $M_S 7.0/M_W 6.6$ earthquake occurred in Sichuan province. The isoseismals (Fig. 9.2) and the losses (Table 9.3) have been published by the CEA (see *Data and Resources*). The isoseismals with $I_{\max} = IX$ are elongated in a NEN direction. The State Council of China released the following loss report on 24 April 2013: 217 deaths (196 deaths and 21 missing) and at least 13,484 injured.

According to the slip model of Zhang et al (2013), we used rupture end-points which are 30.14° N, 102.93° E and 30.35° N, 103.01° E to define a line source. The source properties are: length of 25 km, width of 30 km and dip of 30°. This calculation of casualties shows the smallest differences from the reported casualties when we used a depth of 26 km along with $C_3 = 4.0$. The distance by which we moved the epicenter from the CENC to the macroepicenter is about 6 km. The estimated number of fatalities and injured are 210–3000 and 1000–11,000, respectively (Table 9.3).

QLARM operators do not calculate damage and casualties for regular aftershocks of large earthquakes because the main shock has changed the properties of buildings in the area by eliminating weak ones (collapse) and by weakening strong ones (damage). In addition, people are likely staying in tents, which means the occupancy rate is different than normal. Therefore, calculations of consequences of aftershocks are meaningless. People who distribute such calculations based on automatic estimations are misleading the local population and rescuers.

However, the Lushan case is different from a regular aftershock. We introduce the term “aftershock of the second kind” for loss risk estimates of the Lushan earthquake

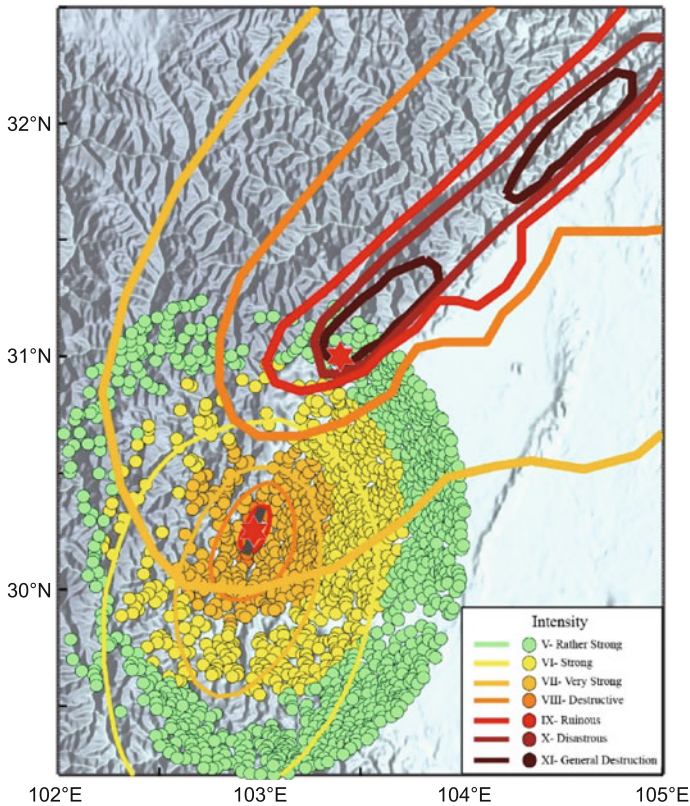


Fig. 9.2 Map of isoseismals of the Lushan earthquake (thin lines) and the Wenchuan earthquake (thick lines) compared with intensities calculated by QLARM (colored dots). The hexagrams indicate the macroscopic epicenter for the Lushan and the instrumental one for the Wenchuan earthquake. The gray solid line depicts the line source we used for calculating intensities for the Lushan earthquake. Parameters used by QLARM for the calculations are given in Tables 9.1 and 9.2.

because its area of strong shaking encroaches on that of the Wenchuan earthquake that happened five years earlier (Fig. 9.2). After the Wenchuan disaster, many new resistant buildings were constructed, and damaged buildings were retrofitted. This increase of building strength five years after the Wenchuan earthquake is opposite to the decrease of resistance in unmodified buildings in days and weeks immediately following a large earthquake. The losses caused by the Lushan earthquake in the area where the building stock has been strengthened cannot be calculated using the regular vulnerability model for the region. The losses in the Lushan earthquake depended on the changes to the building stock after the Wenchuan earthquake. This means that we should not use this earthquake as a validation case.

We observe that fewer fatalities, but more injuries than those expected for the Lushan earthquake were reported. In the near real-time sense, however, the calculation of QLARM was still approximately correct. During the Lushan earthquake, the

southern part where buildings had remained unchanged, were affected severely. This part produced the usual number of casualties and injured. In contrast, in the northern part where buildings are now stronger, thus a portion of the expected fatalities were shifted into the category of “injured”. Therefore, the observed number of injured was larger than expected in the near real-time calculation, and the observed fatalities were less than the average expected (Table 9.3).

9.3 Summary of Validation

For validation of theoretical casualties, the large magnitude earthquakes with numerous casualties are the most important ones, as they are also the most important ones for first responders and for mitigation plans. Matches for earthquakes with few casualties are subject to large variations because the collapse of a single apartment building, school or office building can cause in the order of 200 casualties. In 2008, for example, during the Wenchuan earthquake, main teaching building of Beichuan middle school of Beichuan county collapsed during the school hours. There were about 1,000 teachers and students in 21 classrooms, except for a few who escaped, most were buried under the rubble. Therefore, an over-estimation of theoretical casualties by 200 is acceptable for small events, for which near zero casualties have been reported.

For the above mentioned reasons, we give a high weight for calibration events with significant numbers of casualties, whereas a low weight is given for cases with small numbers of casualties. We conclude that the matches for the three events with large numbers of casualties are within a factor of 1.1 of the reported values (Fig. 9.1, Table 9.3). These excellent matches include the Wenchuan, the Yushu, and the Ludian earthquakes (Table 9.3). The details of 10 validation matches of calculated with observed casualties and isoseismals are discussed in detail in the section of *Supplemental Material*.

The overall conclusion mainly based on the large catastrophes for which QLARM works correctly: is that the QLARM program for estimating casualties in future scenario earthquakes is valid for the Chinese mainland.

Five cases among the validations are not far from the target region, in which we estimate the consequences of scenario earthquakes. They are the Dayao ($C_3 = 3.5$), Wenchuan ($C_3 = 3.7$), Lushan ($C_3 = 4.0$), Ludian ($C_3 = 4.1$) and Jinggu ($C_3 = 4.1$) earthquakes. So, we propose to use the average attenuation that fits best the validation cases, namely $C_3 = 3.9$, in the scenario estimations conducted subsequently.

9.3.1 Near Real-Time Earthquake Loss Alerts for the Mainland of China

Some near real-time loss estimations for the earthquakes of China's mainland have been released by the World Agency for Planetary Monitoring and Earthquake Risk Reduction (WAPMERR, Geneva, Switzerland) and the International Centre for Earth Simulation Foundation (ICES Foundation, Geneva, Switzerland; see *Data and Resources*) since QLARM was used to calculate casualties and distribute loss alerts starting in October 2002. The average delay time between the origin time of the earthquake and the loss alerts is 29 min, which depends in part on the time that USGS, GFZ (the Geoforschungszentrum) and EMSC (European Mediterranean Seismological Center) release the earthquake source parameters because all alerts based on QLARM are calculated and released manually (Wyss, 2014).

Precise parameters of the fault which caused the earthquake are not available in the first minutes after the event. For instance, the azimuth of the fault rupture is not known at that time, and therefore point source approximations are used for near real-time loss calculations. Also, epicenters available in near real-time are usually incorrect with errors of at least 10 km, and the depths is not well known. The macroseismic distribution is especially informative, but it is not known within weeks and sometimes months after earthquakes. All of this renders the near real-time loss estimations of QLARM strongly dependent on uncertainties. Therefore differences between rapidly calculated and final reported casualties must be expected.

However, we show that all the near real-time fatality estimates for mainland of China are correct. For estimates of the numbers of injured, six cases are hundred percent correct and three cases are within a factor of 2.5. Table 9.3 shows the loss alerts calculated by QLARM for the mainland of China and distributed by WAPMERR and ICES Foundation within the period from 1 January 2003 to 1 April 2017, along with the calculated results of this paper and the reported casualties. For small numbers, a difference less than 200 is not wrong if we take into account the collapse of one single building that can kill several hundred, or more people, as seen in Beichuan in the 2008 Wenchuan earthquake.

9.4 Earthquake Scenario Casualty Calculations

Predicting earthquakes, is extremely difficult because many factors, may influence the occurrence time and magnitude. For a valid earthquake prediction, origin time, magnitude and location should be specified with uncertainties, and in addition, the probability of the event to occur should be given.

Here, we do not predict future earthquakes. Instead, we identify active faults that have generated large earthquakes in the past, and which will rupture again in the future. We do not know when they will rupture, but suggest strongly that residents

and authorities should prepare for these earthquakes, which could happen in their respective lifetimes or that of the next generations.

We model only the largest ruptures one can expect, according to the length of the investigated fault segments. Although small and even medium earthquakes occurring frequently along some of them may generate losses that are tragic, here we focus only on large earthquakes, which would be of national important disaster.

We assume 15 km to be the depth of energy release, because the final results from validations tell us that 15 km is an appropriate choice for modelling the main release of energy, on average.

9.4.1 Reasons for Selecting Scenario Earthquake Source Parameters

We selected the Xianshuihe, Anninghe, Zemuhe, Daliangshan and Xiaojiang faults for earthquake scenario calculations (Fig. 9.3). This choice is mainly based on the following three considerations:

- (1) These faults are included in the National Earthquake Monitoring and Prediction Experiment Site of Sichuan-Yunnan (23° – 32° N, 98.5° – 104.5° E; see *Data and Resources*) and China Seismic Experimental Site (21° – 32° N, 97.5° – 105.5° E; see *Data and Resources*);
- (2) The scenarios are located in the southern part of the central China north–south seismic belt, which has been selected to be one of the testing regions of the Collaboratory for the Study of Earthquake Predictability (CSEP) project (Mignan et al., 2013; Zhang et al., 2016a, 2016b);
- (3) This choice is based on outcomes of the annual consultation meetings of the CEA, which has focused on this area in recent years.

We also consulted the fifth-generation seismic hazard zoning map of China, see *Data and Resources*. The peak ground acceleration with 10% probability of exceedance in 50 years is 0.2–0.4 g along the scenario faults and in surrounding regions. The displacement accumulation rates are 10–20, 6–7, 6–7, 3–4 and 10 mm/yr for the Xianshuihe, Anninghe, Zemuhe, Daliangshan and Xiaojiang faults, respectively (Xu et al., 2003, 2005). The assumed source parameters of these faults to rupture in the future possible earthquakes are shown in Table 9.5.

Frequently earthquakes rupture the Xianshuihe fault. At least eight $M > 7.0$ earthquakes have occurred there since the year 1725 (Fig. 9.3). We measure the total length of the Xianshuihe fault as being about 320 km (Xu et al., 2003, 2005), with an assumed width of 20 km (Fig. 9.3, Table 9.5).

The Anninghe fault starts at Shimian in the north and ends at Xichang in the south (Fig. 9.3, Table 9.5). Ran et al. (2008) investigated the recurrence interval of historical major earthquakes and suggested that the Anninghe fault alternates between strong and weak activity. We measure the length of this fault as 140 km (Xu et al., 2003, 2005), the width of the fault plane is assumed to be 20 km (Fig. 9.3, Table 9.5).

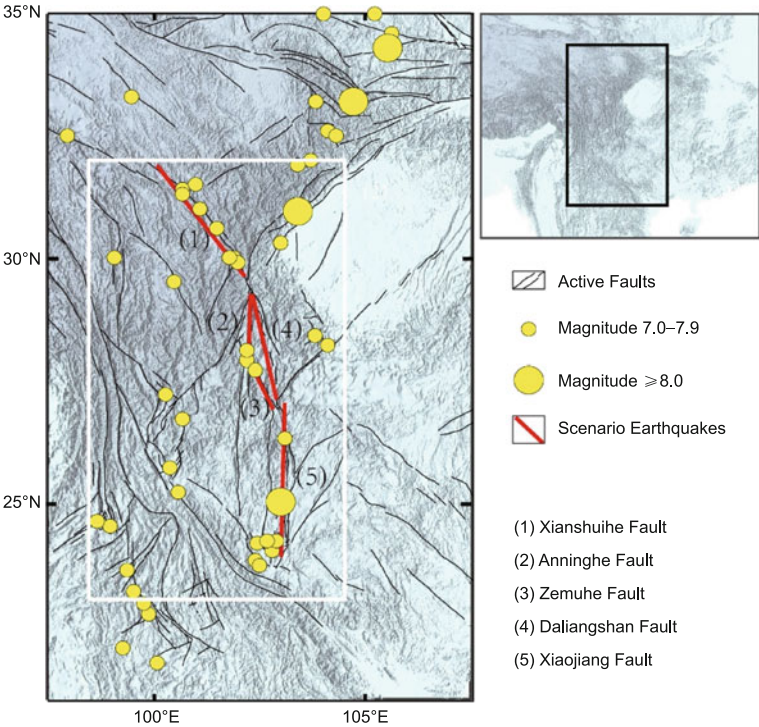


Fig. 9.3 Map of the scenario ruptures (heavy red lines), historical earthquakes (yellow dots) (Department of Earthquake Disaster Prevention of the China Earthquake Administration, 1999; Department of Earthquake Disaster Prevention of the State Seismological Bureau, 1995) on record from 0 BC to AD 2017 with $M \geq 7.0$ and active faults (Deng et al., 2003) in the target region (21.0°–35.0° N, 97.5°–107.5° E). The white rectangle shows the region of the National Earthquake Monitoring and Prediction Experiment Site of Sichuan-Yunnan (23°–32° N, 98.5°–104.5°E). The indexing figure at the top right shows the region under discussion

Table 9.5 Source parameters of the hypothetical earthquakes

Fault	Epi Lat.	Epi Lon.	Z	L	W	Dip.	M(W&C)	Y ₁	X ₁	Y ₂	X ₂
Xianshuihe	30.75	101.13	15	320	20	70	7.9	31.89	100.10	29.61	102.16
Anninghe	28.60	102.27	15	140	20	70	7.5	29.25	102.29	27.95	102.25
Zemuhe	27.43	102.53	15	130	20	60	7.5	27.95	102.23	26.90	102.82
Daliangshan	28.18	102.62	15	250	20	60	7.8	29.25	102.33	27.10	102.90
Xiaojiang	25.45	103.05	15	350	20	70	7.9	27.03	103.09	23.87	103.01

X_i and Y_i are the assumed end points of the respective line ruptures. M is calculated from the rupture area. The epicenter is placed at the center of the line rupture

The Zemuhe fault links with the Anninghe fault in the Xichang region and extends to Qiaojia (Fig. 9.3, Table 9.5). We measure its length as 130 km (Xu et al., 2003, 2005) and assume that the width is 20 km (Fig. 9.3, Table 9.5).

The Daliangshan fault starts at Shimian, and extends to Ningnan (Fig. 9.3, Table 9.5). According to the Chinese historical literature since 0 BC, no large earthquake has occurred along this fault. However, paleoseismological excavations of old slips suggests that there occurred at least nine $M > 7$ earthquakes before the advent of written records (Song et al., 2002). We measure its length as being 250 km (Xu et al., 2003, 2005), while its width is assumed to be 20 km (Fig. 9.3, Table 9.5).

The northern segment of the Xiaojiang fault links with the Zemuhe fault (Fig. 9.3, Table 9.5). The Songming $M 8.0$ earthquake is the largest earthquake along the Xiaojiang fault since 0 BC. It occurred in 1833 and caused severe damage. We measure its total length as being about 350 km (Xu et al., 2003, 2005); the assumed width is 20 km (Fig. 9.3, Table 9.5).

9.4.2 *Approximate Results of Casualty Scenarios*

The estimated maximum expected magnitude for the Xianshuihe rupture from fault geometry is 7.9 (Wells & Coppersmith, 1994, Table 9.5). For the scenario of the Xianshuihe fault, we calculated the numbers of possible fatalities as 68,000-167,000, and injured as 181,000-452,000 (Table 9.6). We assume the occurrence time as 19:00 UTC, which equals 03:00 local time, and which therefore means that the occupancy of buildings is at a maximum. The approximate numbers of people in the area of minor damage (VI + VII) and in the devastated area (VIII+) are 19,860,000 and 5,955,000, respectively (Table 9.7). We defined the sum of these numbers as the strongly affected population.

The estimated maximum expected magnitude for the Anninghe rupture is 7.5, based on the assumed rupture area (Wells & Coppersmith, 1994). For the scenario of a rupture with $M = 7.5$ of the Anninghe fault along with a maximum occupancy of houses, our calculations show that the possible fatalities and injured are 49,000-134,000 and 131,000-364,000, respectively (Table 9.6). The approximate numbers of people in the intensity VI + VII and VIII+ areas are 16,253,000 and 5,348,000 respectively (Table 9.7).

The estimated maximum expected magnitude for the Zemuhe rupture is 7.5, based on the assumed rupture area (Wells & Coppersmith, 1994). With this earthquake occurring at 03:00 local time, the range of numbers of fatalities and injured estimated by QLARM are 38,000-111,000 and 104,000-302,000, respectively (Table 9.6). The approximate numbers of the population in the area of minor damage and in the devastated area are 14,350,000 and 4,376,000, respectively (Table 9.7).

The estimated maximum expected magnitude for the Daliangshan rupture is 7.8, according to the rupture area we specified (Wells & Coppersmith, 1994). For this earthquake occurring at 19:00 UTC the results calculated by QLARM show that the possible numbers of fatalities and injured are 111,000-259,000 and 292,000-693,000,

Table 9.6 The range of numbers of casualty estimated by QLARM for each earthquake scenario

Fault	Xianshuihe	Anninghe	Zemuhe	Daliangshan	Xiaojiang
Fatality	68,000–167,000	49,000–134,000	38,000–111,000	111,000–259,000	176,000–400,000
Injured	181,000–452,000	131,000–364,000	104,000–304,000	292,000–693,000	459,000–1,053,000

Table 9.7 The approximate numbers of people in the areas of intensities (VI + VII) and (VIII+) level in each earthquake scenario, the sum of these numbers are the strongly affected people

	Xianshuihe	Anninghe	Zemuhe	Daliangshan	Xiaojiang
People in the minor damage area (VI + VII)	19,860,000	16,253,000	14,530,000	27,047,000	24,433,000
People in the devastated area (VIII+)	5,995,000	5,348,000	4,376,000	9,258,000	16,361,000

respectively (Table 9.6). The approximate numbers of people in the area of minor damage and in the devastated area are 27,047,000 and 9,258,000, respectively (Table 9.7).

The estimated maximum expected magnitude for the Xiaojiang rupture from fault geometry is 7.9 (Wells & Coppersmith, 1994). For this earthquake, we calculated the possible numbers of fatalities as 176,000-400,000 and injured as 459,000-1,053,000 (Table 9.6) at 19:00 UTC, when there would be a maximum occupancy of dwellings. The approximate numbers of people in the area of minor damage and in the devastated area are 24,433,000 and 16,361,000, respectively (Table 9.7).

9.5 Discussion

Scenario analyses could be done in a more useful way if we had more detailed data on buildings and their uses. For example, we should have a separate data set of locations and sizes of schools and another data set of health facilities, to estimate their probable damage in future earthquakes. This would help with both decision-making for reinforcement and for first responders (e.g. the rescue team). We have proposed to accumulate these data by an Open Building Map (OBM) project (Danijel Schorlemmer, personal communication; the OBM project see Data and Resources). In addition, it would be helpful to construct more detailed and more useful city models.

There are several source properties that influence damage and casualty numbers, which are currently not modeled. The impact of source complexity on estimation of earthquake casualties is not possible to be evaluated at this stage of development of alert tools. Using line sources for validations and scenarios is the best that can be done at this time, and it yields more accurate results than point sources (Li et al., 2020).

The spectrum of energy radiated by a rupture is unknown and not modeled, although it can influence the response. Earthquakes that cause more casualties radiate more than average energy at high frequencies, as suggested in the Ludian and Jinggu cases described above. The moment magnitude M_w is a low frequency concept, that does not measure the radiation of energy (Song et al., 2016), it would be desirable to estimate the impact of source spectra.

Earthquakes in high stress environments such, as thrusts, may cause more casualties than low stress events along normal faults, but this has not been investigated. In addition, the flip effect on the hanging-wall of thrust faults may augment accelerations (Brune, 1996, 2001a, 2001b).

Earthquakes with super-shear ruptures can lead to more serious disasters than normal earthquakes (Das, 2007). Analyses of past disasters indicate that super-shear ruptures can occur on strike slip faults (Xu et al., 2015; Zhang et al., 2017) and the damage caused by earthquakes depends in part on how fast the fault ruptures. The rupture speed is usually not known and it is not modeled in existing earthquake loss estimating tools.

A ubiquitous source of uncertainty is the inaccuracy of the location of epicenters and that of asperities along the fault plane. Epicenter errors in near real-time can map into large errors in casualty estimates (Wyss & Kossobokov, 2012; Wyss et al., 2011). Even final epicenter estimates by regional seismograph networks can deviate from those derived from macroseismic observations by up to 10 km, as shown in this paper. In the case of relatively small earthquakes in areas with uneven population distribution, these uncertainties can lead to significant errors in casualty estimates.

The extent and direction of ruptures could be defined accurately by analyses of satellite images. The use of this technique might dramatically increase the accuracy of near real-time loss estimates, as shown by Wyss et al. (2006). The frequency of satellite passes and the automatization of analysis of satellite data may have moved this approach into the realm of reality.

Local intensities of shaking are not known in near real-time. They are estimated months after an earthquake and are poor approximations to actual ground accelerations. In the future we expect that in a region with a high density MEMS (micro-electro-mechanical systems) for rapid intensity reports, casualties may be estimated more accurately. For example, in the mainland of China after the completion of the National System for Fast Report of Intensities and Earthquake Early Warning project, we will be able to use the near real-time reported intensity distribution to calculate casualties.

The Longmenshan fault zone consists of four parallel faults, including the front-range Longmenshan fault, the central Longmenshan fault, the back-range Longmenshan fault and the mountain front blind fault. The Wenchuan earthquake ruptured on the Beichuan-Yingxiu fault in the middle section of the central Longmenshan fault and the Guanxian-Jiangyou fault in the middle part of the front-range Longmenshan fault. We should not ignore the seismic hazard of the Longmenshan fault zone although the 2008 Wenchuan earthquake (Chen & Booth, 2011) and the 2013 Lushan earthquake (Wu et al., 2014) have recently occurred nearby on it. Assessing the seismic hazard along the Longmenshan fault zone, Wang et al. (2010) considered seismic moment deficits (unreleased moment) or surplus (over-released moment) and found that the Wenchuan earthquake did not release all seismic moment on the fault. However, the buildings that would be affected in a Longmenshan scenario are probably mostly those that have been rebuilt after the 2008 Wenchuan earthquake and are stronger than they were before. Therefore, we do not have the correct building stock

for this scenario. In the future, the losses should be further studied for this case, with a new model used for the building stock.

The Yutian $M_S7.3/M_W6.8$ earthquake of 12 February 2014 in Xinjiang Uygur Autonomous Region does not qualify for a calibration case. The fact that nearly all calculations result in 0 or very few fatalities is due to the absence of people in the affected region. There is a settlement in the field of $I = IX$ and another settlement at the edge of the fields VIII and VII. Our calculations depend on many settlements contributing to the loss because the soil conditions and other variations are approximately averaging out. We cannot rely on the Yutian event with near zero input as a pillar for documenting the performance of QLARM, thus we did not use it.

9.6 Engineering Measures to Reduce Casualties

Although we are not specialists in engineering, a few remarks about how to reduce casualties are offered in this article. There exists a great opportunity to save lives by strengthening buildings. Local standards for seismic precautionary intensity should be reassessed and raised to a new reasonable level appropriate for buildings to withstand the shaking of intensities VII–IX.

Some potentially dangerous houses, i.e. those constructed of brick and adobe, should be reinforced with lateral resisting systems (e.g. shear walls, frames with in-fill walls and constructional columns), or reconstructed with reinforced concrete materials. The reinforced concrete structures should obey the principle of “strong column and weak beam” and, for those buildings which already exist, but do not meet the earthquake resistance level, carbon fiber sheet and steel-bonded reinforcement technique should be introduced (Hu, 2009).

If we assume that 70–90% of fatalities and injured could be saved by strengthening buildings, this would mean that the number of fatalities and injured could be reduced by 30,000–350,000 and 70,000–940,000, respectively, in each earthquake, depending on the particular case. This assumption of a reduced casualty rate is realistic. Xinjiang Uygur Autonomous Region of China is an earthquake-prone region. The earthquake of $M6.8$ occurred in the Bachu-Jiashi region on February 24, 2003, and killed 268 and injured 4,853. Since 2004, the local government has launched a rural earthquake-resistant project. More than 50 earthquakes larger than $M5$ occurred in Xinjiang during the 10 years from 2004 to 2014, but no one died and only 73 people were injured.

In addition, for people who cannot afford retrofitting the whole house, an earthquake closet that could improve the resident’s chance of survival and not being injured by factors of 1000–30,000 (Wyss, 2012) is recommended. This factor depends on the ratio of the resistances to shaking by the building to that of the earthquake protection unit (EPU). For a house built poorly of heavy material that will collapse, the improvement of surviving in an EPU that is constructed as strong as a bunker, is almost infinity. In case of strong shaking, a family has about 5–10 s time to get into

the EPU because this time span is equal to the difference between the first arriving P-wave with small amplitudes, and the later-arriving, destructive S-wave.

9.7 Conclusions

We validate the algorithms and data sets of QLARM for the 13 earthquakes of the mainland of China since 2003 for which intensity and fatality information exist (Fig. 9.1). It appears isoseismals are more accurate than instrumental locations for pinpointing the centroid. The distance on average from the instrumental epicenter to the macroepicenter is about 9 km for these validations.

Considering an earthquake with small casualty numbers, a loss difference less than 200 is acceptable because the collapse of one single building can kill many people, or more than the expected death toll. Therefore, we concluded that 13 near real-time fatality estimates are correct. For the results of injured, six cases are 100% correct and three cases are within a factor of 2.5.

For the three events with large numbers of casualties (the Wenchuan, the Yushu, the Ludian earthquakes) to which we give the most weight, the matches of theoretical intensities and numbers of the casualties are within a factor of 1.1 of the reported values (Fig. 9.1, Table 9.3).

Although several improvements in loss estimates may become available in the future, the near real-time verification calculations reported herein show that earthquake losses can be estimated within factors of 2–3 in the mainland of China using the currently available methods and data sets. It gives us confidence that the QLARM program for estimating casualties in future scenario earthquakes is applicable for the mainland of China.

We select the Xianshuihe, Anninghe, Zemuhe, Daliangshan and Xiaojiang faults (Fig. 9.3) for possible ruptures in their entire length in the Sichuan-Yunnan region, China, and construct “worst-case scenarios” of devastating earthquakes. We assume the occurrence time at 03:00 local time, which implies maximum occupancy of buildings. For the five scenarios, the numbers of possible casualties are calculated and listed in Table 9.6.

The greatest disaster is expected for the scenario of the Xiaojiang fault for which we calculated the possible numbers of fatalities on average as being 286,000 and injured as being 751,000. These expected numbers of casualties are about a factor of 3 larger than those reported in the Wenchuan earthquake of 2008. Part of the reason for this is that we assume better transmission of seismic waves in the Xiaojiang scenario cases than in Wenchuan.

The scenario of the maximum rupture along the Daliangshan fault results in approximately twice the casualties as reported in the Wenchuan earthquake. The three scenarios along the Xianshuihe, Anninghe and Zemuhe faults result in estimates of casualties similar to those in the Wenchuan case. Although our calculations are approximate, they are realistic. Earthquakes similar to the ones modeled here are inevitable, only their occurrence time is unknown. We hope that the numbers of

casualties we forecast here motivate the government and the local inhabitants to take preventive measures.

9.8 Data and Resources

The earthquake catalog of USGS was searched via <https://earthquake.usgs.gov/earthquakes/search/> (last accessed on 20 September 2017). The annual population growth rate for China is taken from the National Bureau of Statistics of China, available at <http://data.stats.gov.cn/search.htm> (last accessed on 21 September 2017). The different rupture models of the 2008 Wenchuan earthquake of China can be also found via the finite-source rupture model database (<http://equakerc.info/SRCMOD/searchmodels/allevnts/> , last accessed 21 September 2017) and the U.S. Geological Survey (<https://earthquake.usgs.gov/earthquakes/eventpage/usp000g650#finite-fault>, last accessed 21 September 2017). The rupture model of the 2016 Arketao earthquake are taken from the Institute of Geophysics, China Earthquake Administration, available at <http://www.cea-igp.ac.cn/tpxw/275080.html> (last accessed on 20 September 2017). The reported numbers and isoseismals for Chinese earthquakes can be found at the disaster assessment notice of the website of CEA (<http://www.cea.gov.cn/publish/dizhenj/468/549/index.html>, last accessed on 21 September 2017). The alerts released by the International Centre for Earth Simulation Foundation can be accessed at <http://www.icesfoundation.org/Pages/CustomPage.aspx?ID=122> (last accessed 21 September 2017). The information about the National Earthquake Monitoring and Prediction Experiment Site of Sichuan-Yunnan can be accessed at <http://www.ncepe.cn/syccn/index.php> (last accessed 24 September 2017). The new generation (fifth generation) seismic hazard zoning map of China can be found at <http://www.gb18306.cn/> (last accessed 22 September 2017). The information of Open Building Map (OBM) project are accessed at <http://www.openbuildingmap.org> (last accessed 12 January 2018). The news of the rural earthquake-resistant project of Xinjiang can be found at <http://www.chinanews.com/sh/2014/02-14/5838784.shtml> (last accessed 2 April 2018).

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